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INVITED TALK / ABSTRACT

When Mean Curvature Flow Breaks: Singularities, Topology, and Ilmanen's Conjecture

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Abstract

Mean curvature flow evolves a surface in the direction that most rapidly decreases area. Although the flow tends to smooth geometry, singularities inevitably form. A fundamental question is how the topology of the surface behaves at such singular times. Ilmanen conjectured that mean curvature flow should not create new genus. In this talk I will describe the background of this problem and explain recent joint work with David Hoffman and Brian White resolving the conjecture. The examples show that singularities in mean curvature flow can be far richer than expected.

Keywords mean curvature flow; singularities; topology